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Investigation of practical approaches to evaluating cumulative dose for cone beam computed tomography (CBCT) from standard CT dosimetry measurements: a Monte Carlo study

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Abstract

A function called $G_x(L)$ was introduced by the International Commission on Radiation Units and Measurements (ICRU) Report–87 to facilitate measurement of cumulative dose for CT scans within long phantoms as recommended by the American Association of Physicists in Medicine (AAPM) TG–111. The $G_x(L)$ function is equal to the ratio of the cumulative dose at the middle of a CT scan to the volume weighted CTDI ($CTDI_{vol}$), and was investigated for conventional multi-slice CT scanners operating with a moving table. As the stationary table mode, which is the basis for cone beam CT (CBCT) scans, differs from that used for conventional CT scans, the aim of this study was to investigate the extension of the $G_x(L)$ function to CBCT scans. An On-Board Imager (OBI) system integrated with a TrueBeam linac was simulated with Monte Carlo EGSnrc/BEAMnrc, and the absorbed dose was calculated within PMMA, polyethylene (PE), and water head and body phantoms using EGSnrc/DOSXYZnrc, where the body PE body phantom emulated the ICRU/AAPM phantom. Beams of width 40–500 mm and beam qualities at tube potentials of 80–140 kV were studied. Application of a modified function of beam width (W) termed $G_x(W)$, for which the cumulative dose for CBCT scans $f(0)$ is normalized to the weighted CTDI ($CTDI_w$) for a reference beam of width 40 mm, was investigated as a possible option. However, differences were found in $G_x(W)$ with tube potential, especially for body phantoms, and these were considered to be due to differences in geometry between wide beams used for CBCT scans and those for conventional CT. Therefore, a modified function $G_x(W)_{100}$ has been proposed, taking the form of values of $f(0)$ at each position in a long phantom, normalized with respect to dose indices $f_{100}(150)_x$ measured with a 100 mm pencil ionization chamber within standard 150 mm PMMA phantoms, using the same scanning parameters, beam widths and positions within the phantom. $f_{100}(150)_x$ averages the dose resulting from a CBCT scan over the 100 mm length. Like the $G_x(L)$ function, the $G_x(W)_{100}$ function showed only a weak dependency on tube potential at most positions for the phantoms studied. The results were fitted to polynomial equations from which $f(0)$ within the longer PMMA, PE, or water phantoms can be evaluated from measurements of $f_{100}(150)_x$. Comparisons with other studies, suggest that these functions may be suitable for application to any CT or CBCT scan acquired with stationary table mode.

Keywords: cumulative dose, CBCT, $CTDI_{100}$, EGSnrc/BEAMnrc, EGSnrc/DOSXYZnrc, AAPM TG-111, ICRU Report-87

(Some figures may appear in colour only in the online journal)

1. Introduction

The importance of computed tomography (CT) scanning for medical imaging has increased steadily over the last four decades. The rapid development in CT technology in recent years has broadened the areas of application. For example, the introduction of flat panel detectors for digital radiography opened up the possibility of obtaining a CT scan from a single rotation of the x-ray tube using a cone beam, and this has subsequently been employed for various cone beam CT (CBCT) scan applications such as image guided radiation therapy (IGRT) and interventional procedures. As the ionizing radiation involved in CT and CBCT scans may have an associated health detriment (Brenner and Hall 2007, Pearce *et al* 2012), it is necessary to optimise the dose required for the scans and consider the trade-off between the benefits and risks (AAPM 2007). Therefore, it is important to have effective ways for evaluating doses from CT and CBCT scans. The dose descriptor that has been employed for CT dosimetry for many years is known as the CT dose index (CTDI) (Shope *et al* 1981), which is based on integrating the axial dose profile resulting from a single axial rotation over an arbitrary length at the middle of the scan ($z = 0$). The CTDI has been adapted several times to accommodate advances in CT scanner technology. Different derivatives of the CTDI concept have been developed based on the CTDI measured with a 100 mm long pencil ionization chamber known as ($CTDI_{100}$). The $CTDI_{100}$ is the main index from which other derivatives are derived such as the weighted CTDI ($CTDI_w$), volume weighted CTDI ($CTDI_{vol}$), dose-length product (DLP) and $CTDI_{100}$ for CBCT scans (IEC 2001, McNitt-Gray 2002, IEC 2010, IAEA 2011, Kalender 2014). The $CTDI_{100}$ can be used to characterize the output of a CT scanner by making measurements free in air at the isocentre, and parallel to the scanner axis at a specific source isocentre distance (SID). For assessment of the doses received by patients undergoing CT scans, the $CTDI_{100}$ is measured in cylindrical phantoms made of polymethyl methacrylate (PMMA), 320 mm or 160 mm in diameter and 150 mm in length placed at the SID.

With the continued evolution of CT scanner technology, preservation of the $CTDI_{100}$ as a dose index is problematic, as it can only be utilized for a limited range of beam widths, and in CBCT, the beam is often wider than the chamber length of 100 mm (Dixon 2003, Brenner 2005, Mori *et al* 2005, Dixon and Ballard 2007, Boone 2007, Kyriakou *et al* 2008, Boone 2009). Moreover, the efficiency of the $CTDI_{100}$, which is the ratio of the $CTDI_{100}$ parameter measured in a PMMA phantom of standard length to the equivalent $CTDI_{\infty}$ parameter measured in an infinitely long phantom, is limited even for narrow beams. The $CTDI_{\infty}$ includes the dose from all the scattered radiation generated within a phantom undergoing a CT or CBCT scan that reaches the measurement chamber, and will be closer to the radiation level from a scan within a human trunk. The efficiency values are approximately constant for beams of width ≤40 mm at ~75 and ~60% at the centre of the standard PMMA head and body phantoms, respectively, and at ~84% at the periphery of the phantoms (Dixon and Ballard 2007, Perisinakis *et al* 2007, Ruan *et al* 2010, Martin *et al* 2011, Li *et al* 2011, Li *et al* 2012). The efficiency values, however, decrease with increasing beam width reaching ~25% of $CTDI_{\infty}$ for wide beams for both the head and body phantoms (Boone 2007, Kyriakou *et al* 2008, Abuhaimed *et al* 2014).

In order to avoid the underestimation of dose associated with the use of $CTDI_{100}$ for wide beams, such as CBCT scans, various practical approaches have been suggested (Mori *et al* 2005, Islam *et al* 2006, Fahrig *et al* 2006, Amer *et al* 2007, Kyriakou *et al* 2008, Geleijns *et al* 2009, IEC 2010, AAPM 2010, Dixon and Boone 2010). These practical approaches were investigated in detail in previous studies (Abuhaimed *et al* 2014, Abuhaimed *et al* 2015). Two approaches have been proposed by international organisations. One by the International Electrotechnical Commission (IEC) (IEC 2010) is based on modifications to measurement of the $CTDI_{100}$, and this has been recommended by the International Atomic Energy Agency (IAEA) (IAEA 2011) and the Institute of Physics and Engineering in Medicine (IPEM) (Platten *et al* 2013). However, the methodology proposed by the American Association of Physicists in Medicine (AAPM) Task Group—111 (AAPM 2010) aimed to replace the $CTDI_{100}$ with a concept based on the cumulative dose that requires measurements using a chamber of a small active detection length under a scatter equilibrium condition. This equilibrium condition is achieved by using a cylindrical phantom made of PMMA, polyethylene (PE), or water that has a length of at least 450 mm as shown in figure 1. The use of such large and heavy phantoms in the clinical environment for routine quality assurance (QA) measurements is impractical. Therefore, different practical methods have been investigated to overcome this difficulty.

The International Commission on Radiation Units and Measurements (ICRU) Report–87 proposed a practical approach to utilize the AAPM method, but avoid the difficulty of using the long phantoms (ICRU 2012). This approach is based on the application of a function called $G_x(L)$. Moreover, the ICRU in cooperation with the AAPM Task Group—200 (AAPM 2015) introduced a new cylindrical phantom named the ICRU/AAPM phantom within which the $G_x(L)$ function is measured. This phantom is shown in figure 1(a), and is designed to emulate an adult body.

Li *et al* (2013) proposed a practical approach to avoid the use of long phantoms, which is suitable for stationary or moving table multi-slice CT (MSCT) scans with beams of width ≤40 mm. This approach is based on the use of $CTDI_{100}$, the efficiency of $CTDI_{100}$, and a function known as the approach to equilibrium function. Li *et al* (2014) also proposed another practical approach for the stationary table mode employed for CBCT scans with beams of width 30–250 mm. This requires measurement of the cumulative dose using a small ionization chamber or a standard 100 mm long pencil ionization chamber within the standard PMMA phantoms and the application of correction factors equal to the ratios of the cumulative doses within infinitely long PMMA phantoms to those within the standard phantoms.

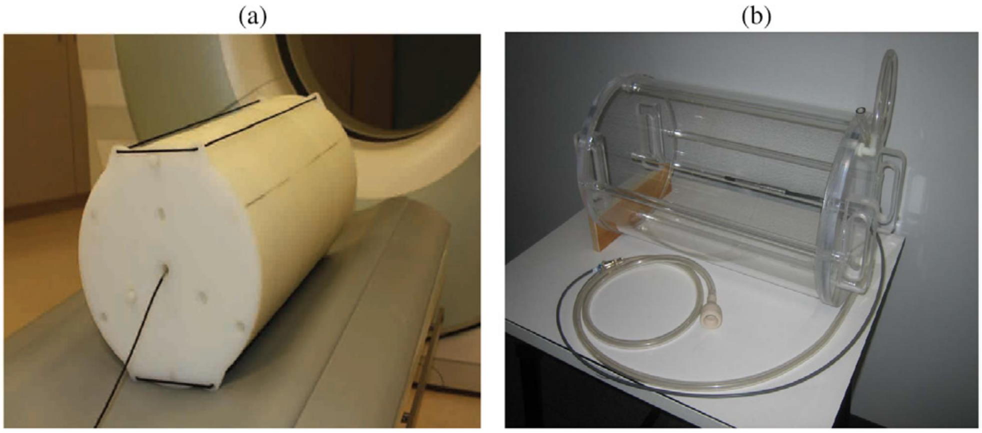

Figure 1. (a) The ICRU/AAPM phantom, made of polyethylene, 300 mm in diameter and 600 mm in length. (b) The water filled phantom recommended by (AAPM 2010) TG—111, 300 mm in diameter and 500 mm in length. Both the phantoms represent an adult body. (a) Reprinted with a permission of John Boone, copyright 2010. (b) reprinted with permission of AAPM and Robert Dixon, copyright 2010.

Dixon and Boone (2014) also proposed two methods to address the long phantom issue for scans obtained with the stationary table mode. The first method requires a single measurement of cumulative dose for a beam of full width at half maximum (FWHM) >24 mm within a long phantom as described in (AAPM 2010) TG—111. Subsequently, the cumulative dose for any beam width of interest can be evaluated by multiplying the cumulative dose measured by the ratio of the approach to equilibrium functions for the beam width of interest $H(a')$ and that for the measured beam width $H(a)$, where $H(a)$ and $H(a')$ are calculated using theoretical equations based on a mathematical model (Dixon and Boone 2010). The second method is based on the use of an analytical formula to evaluate the cumulative dose for a given scan by calculating the primary beam dose component in CTDI_{100}.

The $G_x(L)$ function was investigated in ICRU Report–87 using MSCT scanners. However, dosimetry of the moving table mode differs slightly from that of the stationary table mode which is the basis for CBCT scans such as those used in IGRT, in perfusion scans, in dental scans, and with interventional radiology and cardiology C-arm equipment. The main purpose of the present study was to investigate the extension of the $G_x(L)$ function to CBCT applications.

2. Materials and methods

2.1. The $G_x(L)$ function

The definitions of the dose indices CTDI_{100}, $\mathrm{CTDI}_{\mathrm{w}}$, $\mathrm{CTDI}_{\mathrm{vol}}$, and DLP are given in detail in Appendix A. Initially, the $G(L)$ function, proposed by ICRU, is defined as the ratio of the cumulative dose $D_L(0)$ at the middle ($z = 0$) of an infinitely long phantom, for a CT scan of length (L), to $\mathrm{CTDI}_{\mathrm{vol}}$ for the scan, as follows (ICRU 2012):

$$G(L) = \frac{D_L(0)}{\mathrm{CTDI}_{\mathrm{vol}}} \tag{1}$$

Experimental measurements of the $G(L)$ function at the central axis of the ICRU/AAPM phantom at 120 kV reported in (ICRU 2012) for three different MSCT scanners showed insignificant variations between the scanners. In addition, nearly identical $G(L)$ functions were obtained for four different tube potentials 80, 100, 120 and 140 kV for one MSCT scanner. The $G(L)$ function, therefore, appeared to be independent of scan parameters such as tube potential, beam width, bowtie filter, SID, scanner table composition, and CT scanner model (ICRU 2012). This independence occurs because the influence of scan parameters is cancelled out by the normalization of $D_L(0)$ with respect to $\mathrm{CTDI_{vol}}$, and as a consequence each phantom will have a unique $G(L)$ function that can be used for any CT scanner depending only on phantom diameter and composition. This feature makes the ICRU approach efficient and practical in the clinical environment. When the $G(L)$ function is known, $D_L(0)$ can be estimated for any scan of specified scan length (L) as $D_L(0) = G(L) \times \mathrm{CTDI_{vol}}$, where the $\mathrm{CTDI_{vol}}$ is displayed on the console.

Although the $G(L)$ function was only investigated at the central axis of the ICRU/AAPM phantom, ICRU suggested that the use of the $G(L)$ function can be extended to the periphery of the phantom and hence the weighted value, and can also be employed with other phantoms of varying composition and diameter such as head or body phantoms made of PMMA or water (ICRU 2012). Therefore, the $G(L)$ function of equation (1) can be redefined as:

$$G_x(L) = \frac{D_L(0)_x}{\mathrm{CTDI_{vol}}} \tag{2}$$

where the x indicates the position of the measurement within a phantom, with x = c for the centre or x = p for the periphery, and x = a for the weighted average measurements in the weighted function $G_\mathrm{a}(L)$.

2.2. Modified function for CBCT scans

ICRU investigated and proposed the $G_x(L)$ function equation (2) for use with MSCT scanners with a moving table. The arrangement for CBCT usually uses a single rotation with a broader beam, a flat panel detector, and a stationary table. Therefore the equivalent function should take the differences between conventional CT and CBCT scanners into account. The cumulative dose for CBCT scans at $z = 0$, which is equivalent to $D_L(0)_x$ in equation (2), is equal to $Nf(0)_x$, where N is the number of the rotations acquired in the CBCT scan, and $f(0)_x$ is the peak value of the dose profile resulting from a single axial rotation (AAPM 2010). $f(0)_x$ is measured as a point dose using a small ionization chamber placed at the middle of the central or peripheral axis of an infinitely long phantom. The standard dose quantity $\mathrm{CTDI_{vol}}$ will be replaced by $\mathrm{CTDI_w}$ in equation (2), since there is no table movement and so no pitch to take into account. The length of the scan L equates to the beam width (W), as this determines the axial extent of the scan along the z-axis i.e. rotation axis (Dixon and Boone 2010). Therefore $G_x(L)$ transforms to a function of beam width (W), i.e. $G_x(W)$, and an equation of similar form to equation (2) can be defined for CBCT scans:

$$G_x(W) = \frac{Nf(0)_x}{\mathrm{CTDI_w}} \tag{3}$$

The symbol (W) for the beam width of a CBCT scan should not be confused with (w) for the weighted CTDI ($\mathrm{CTDI_w}$). Both $D_L(0)_x$ and $f(0)_x$ in the numerators of equations (2) and (3) approach asymptotic values as the scan length (L) and the beam width (W) respectively increase, because contributions from scatter originating further from the measurement point decline exponentially (ICRU 2012, Dixon and Boone 2010, Dixon and Boone 2011). Therefore, $f(0)_x$ in equation (3) can be expressed as $f_W(0)_x$ in a manner similar to that for $D_L(0)_x$, but to avoid confusion with the weighted value (w), $f(0)_x$ notation has been used in this study.

However, the fact that $G_x(L)$ is a function of the scan length, whereas $G_x(W)$ is a function of the beam width means that equations (2) and (3) are fundamentally different. This is because the denominators in the equations differ. $\mathrm{CTDI_{vol}}$ for MSCT scans is independent of scan length L, whereas $\mathrm{CTDI_w}$ for CBCT scans decreases with beam width W. The cumulative dose $D_L(0)_x$ in the $G_x(L)$ function equation (2) is normalized with respect to $\mathrm{CTDI_{vol}}$. So as the scan length (L) increases, $D_L(0)_x$ at the central and peripheral axes also increases, but approaches an equilibrium value at a length known as the equilibrium length (L_{eq}), after which further contributions from scattered radiation to $D_L(0)_x$ become negligible (AAPM 2010). The asymptotic curve that describes the dependence of $D_L(0)_x$ on scan length, reaches 98% of the equilibrium cumulative dose $D_{eq}(0)_x$ at L_{eq}, and is known as the approach to equilibrium function (AAPM 2010). The $G_x(L)$ function has a similar dependence on L to $D_L(0)_x$ as $\mathrm{CTDI_{vol}}$ in the denominator is constant. In contrast to $G_x(L)$, the $G_x(W)$ function in equation (3) increases almost linearly with beam width. Although $f(0)_x$ values approach an equilibrium value in the same manner as $D_L(0)_x$ (Dixon and Boone 2010), they are normalized with respect to the $\mathrm{CTDI_w}$, which decreases with increasing beam width (Boone 2007, Kyriakou *et al* 2008, Abuhaimed *et al* 2014). In order to obtain a function with a similar asymptotic behaviour to $G_x(L)$, $\mathrm{CTDI_w}$ has been measured with a reference beam width ($W \leq 40$ mm) and the $G_x(W)$ function in equation (3) redefined as:

$$G_x(W) = \frac{Nf(0)_x}{\mathrm{CTDI_{w,ref}}} \tag{4}$$

where (ref) indicates the reference beam width used for $\mathrm{CTDI_w}$. In this study, a beam of width 40 mm was used to calculate the $G_x(W)$ functions.

2.3. The OBI system

The beam qualities studied were generated using a kV imaging system known as an On-Board Imager (OBI) system mounted on a Varian TrueBeam linear accelerator (Varian Medical systems, Palo Alto, CA). Detailed description of this system was presented in previous studies (Abuhaimed *et al* 2014, Abuhaimed *et al* 2015). In the clinic, two bowtie filters known as full bowtie and half bowtie filters are utilized to optimize the image quality and minimize the dose delivered to patients undergoing scans. The full bowtie filter is used for head scans or scanning an object with a small diameter, whereas the half bowtie filter is used for larger subjects such the adult trunk. Two scanning protocols were studied: (1) Head: a full 360° scan with the full bowtie filter, and (2) Body: a full 360° scan with the half bowtie filter. Four beam qualities at tube potentials 80, 100, 120 and 140 kV were investigated for each protocol, but 125 kV was used instead of 120 kV for the body protocol as 125 kV is employed in the clinic for body scans. Two pairs of collimator blades known as X and Y pairs are integrated within the system to control the size of the beam. The X pair (X_1 and X_2) is used to set the diameter of the scan, whereas the Y pair (Y_1 and Y_2) controls the beam width (W) along the rotation axis, i.e. z-axis. X_1 and X_2 for both the protocols studied were set to the same sizes as those used in the clinic protocols, 264 and 478 mm at the isocentre for the head and body, respectively. Y_1 and Y_2 were set to generate a wide range of beam widths ranging from 40 to 500 mm with an increment of 20 mm. Therefore, the field sizes investigated for the head protocol were [264 mm × (40–500) mm], and for the body protocol [478 mm × (40–500) mm].

2.4. Monte Carlo simulations

The Monte Carlo (MC) model of the kV system used in this investigation has been validated against different experimental measurements using specifications of the kV system obtained from the manufacturer in previous studies (Abuhaimed *et al* 2014, Abuhaimed *et al* 2015). A number of EGSnrc–based (V4 2.4.0) codes (Kawrakow and Rogers 2000, Kawrakow 2000) namely CAVRZnrc (Rogers *et al* 2013a), BEAMnrc (Rogers *et al* 1995, Rogers *et al* 2013b), and DOSXYZnrc (Walters *et al* 2013) were utilized in the design and validation.

In this study, BEAMnrc was used to simulate the kV source, and DOSXYZnrc was employed to calculate the absorbed dose in phantoms made of PMMA, PE, and water. As the kV source beams delivered to all the phantoms are identical, a library of files known as phase space (PHSP) files was created using BEAMnrc to minimize the time required to accomplish the simulations. The PHSP file records information about all the particles resulting from the simulation run with BEAMnrc. This information includes: the charge, the total energy, the position and direction along the x and y directions, and the weighting for each particle, and the position of the last interaction for photons in the z direction towards the target (Rogers *et al* 1995, Rogers *et al* 2013b). A total of 192 simulations, 96 for each scanning protocol, were run using BEAMnrc at the tube potentials and the field sizes described is section 2.3. A PHSP file for each simulation was stored at a source-surface-distance (SSD) of 75 cm. Subsequently, the PHSP files were used as sources in DOSXYZnrc to run 768 simulations, 384 for each protocol to calculate the $G_x(W)$ functions as in equation (4). Data for the materials of the designed model and the phantoms were generated with PEGS4 code. All the simulations were run in the Scottish Grid Service (ScotGrid) at the University of Glasgow.

2.4.1. The kV source simulation. The kV sources were run with BEAMnrc using (0.85–1×10^9) histories and with (ISOURCE = 10: Parallel Circular Beam Incident from Side) of the code, which generated mono-energetic electron beams with a diameter of 1 mm and energies of 80–140 keV to interact with the anode of the kV system from the side at an angle of 14°. It has been shown that the efficiency of a simulation carried out with a low x-ray energy can be enhanced by up to 5–6 times by using a variance reduction technique known as directional bremsstrahlung splitting (DBS) (Kawrakow *et al* 2004, Mainegra-Hing and Kawrakow 2006). The DBS technique, therefore, was used to optimise the simulations by setting the splitting number of the technique (NBRSP) to 2×10^4. For all the simulations, 0.516 MeV and 0.001 MeV were set as the low energy thresholds for creation of secondary electrons (AE) and photons (AP), respectively, and to the cut-off energies for transport of the electrons (ECUT) and photons (PCUT), respectively (Ding and Munro 2013). The default boundary crossing algorithm (EXACT) and electron step algorithm (PRESTA-II) were used. Spin effect, the electron impact ionization, Rayleigh scattering, and atomic relaxations were included in the simulations (Ding and Coffey 2010). The cross sectional data of the national institute of standards and technology (NIST) was used for bremsstrahlung cross sections, Koch–Motz for bremsstrahlung angular sampling, and NIST XCOM for photon cross sections (Mainegra-Hing and Kawrakow 2006).

2.4.2. Dose calculation for the $G_x(W)$ functions. The PHSP files generated from BEAMnrc were run in DOSXYZnrc using (ISOURCE = 8: Phase-Space Source Incident from Multiple Directions) to calculate the $G_x(W)$ function as in equation (4). Eight phantoms made of PMMA, PE, and water representing head and body phantoms (table 1) were designed in DOSXYZnrc. A voxel size of $0.5 \times 0.5 \times 0.5\,\text{cm}^3$ was used along the central and peripheral

Table 1. Densities, chemical compositions, diameters, and lengths of the PMMA, PE, and water head and body phantoms used in this investigation.

Phantom material	Density (ρ) ($g\,cm^{-3}$)	Chemical composition	Head diameter (mm)	Body diameter (mm)	Phantom length (mm)
PMMA	1.19	$C_5O_2H_8$	160	320	150
					600
PE	0.97	C_2H_4	160	300	600
Water	1.0	H_2O	200	300	600

axes of the phantoms, but different voxel sizes were used at the edge and middle regions. Voxels of sizes $0.5 \times 0.5 \times 0.5\,cm^3$ up to $7.75 \times 7.75 \times 0.5\,cm^3$ were used for the head phantoms, and $0.5 \times 0.5 \times 0.5\,cm^3$ up to $13.25 \times 13.25 \times 0.5\,cm^3$ were used for the body phantoms. The main advantage of varying the voxel size in a homogenous phantom is to reduce the simulation time required to calculate the absorbed dose, which could be reduced by more than a factor of three without affecting the simulation efficiency (Babcock *et al* 2008). The four PMMA phantoms were designed with diameters equal to those of the standard PMMA phantoms, and diameters of the head and body water phantoms were those recommended by AAPM TG—111 (AAPM 2010). It has been shown that $CTDI_\infty$ calculated in a PE phantom of diameter 160 mm is comparable to that of a PMMA phantom of similar diameter (Zhou and Boone 2008), thus this diameter was used for the PE head phantom, whereas the body PE phantom recommended by ICRU/AAPM is 300 mm in diameter (figure 1(a)). The short PMMA phantoms (150 mm) represented the standard phantoms used for dosimetry in hospitals and were used to calculate $CTDI_{100}$ values and hence $CTDI_w$ values. The longer phantoms (600 mm) were considered to emulate the infinitely long phantoms within which $f(0)_x$ values for the different phantom compositions were calculated. Only a single rotation $N = 1$ was used, and the $G_x(W)$ function was evaluated at the middle of the central and peripheral axes of the phantom and for the weighted value. Therefore, the $G_x(W)$ function was calculated for the PMMA, PE and water head and body phantoms using equation (4) as:

$$G_c(W) = \frac{f(0)_{c,m}}{CTDI_{w,40}} \quad G_p(W) = \frac{f(0)_{p,m}}{CTDI_{w,40}} \quad G_a(W) = \frac{f(0)_{w,m}}{CTDI_{w,40}} \tag{5}$$

where m represents the composition of the phantom (PMMA, PE or water) used to calculate $f(0)_x$ values, and $f(0)_w$ is calculated in a manner similar to that used for $CTDI_w$, i.e. $f(0)_w = 1/3\,f(0)_c + 2/3\,f(0)_p$.

The same (AE), (AP), (ECUT) and (PCUT) values, and the cross sectional data involved in BEAMnrc simulations were also used in DOSXYZnrc. In order to achieve a statistical uncertainty of less than 1% for each simulation, ($5–6 \times 10^8$) histories with a photon splitting number of 50 were run. As all the simulations were carried out with homogeneous phantoms, the HOWFARLESS transport algorithm and the default boundary crossing algorithm (PRESTA-I) were employed to minimize the simulation times and to increase the efficiency of the dose calculations for the $G_x(W)$ functions (Walters and Kawrakow 2007). Outputs from the DOSXYZnrc simulations were subsequently analyzed by a MATLAB code built in-house.

2.5. $CTDI_{100,x}$ and $f(0)_{x,m}$ experimental measurements:

$CTDI_{100,x}$ and $f(0)_{x,PMMA}$ were measured experimentally for the head and body scanning protocols used in the clinic. The measurements were made within PMMA head and body phantoms using the head (full bowtie filter, 100 kV) and body (half bowtie filter, 125 kV) protocols and for the clinical beam width of 198 mm. The standard PMMA head and body phantoms were used in $CTDI_{100,\,x}$ measurements, and the $f(0)_{x,\,\mathrm{PMMA}}$ measurements were made within PMMA head and body phantoms of length 450 mm, as this length is sufficient to establish the required scatter equilibrium condition (AAPM 2010). The long head and body PMMA phantoms were created by joining together three standard phantoms. A standard 100 mm pencil ionization chamber (20X6-3CT, Radcal Corporation, US) and a 0.6 cm^3 Farmer–type ionization chamber (10X5-0.6CT, Radcal Corporation, US) were utilized for $CTDI_{100,\,x}$ and $f(0)_{x,\,\mathrm{PMMA}}$ measurements, respectively. Both the ionization chambers were calibrated at a standard dosimetry laboratory for the kV energies. The phantoms were set up with the centre at the isocentre with a SID of 100 cm, and the ionization chambers were positioned at the middle of the central and four peripheral axes of the phantoms for the respective measurements.

3. Results and discussion

3.1. Study of the $G_x(W)_m$ function as dosimetry variable for CBCT assessment

Figures (2)–(4) show the $G_c(W)$, $G_p(W)$, and $G_a(W)$ functions for the PMMA, PE, and water head and body phantoms for CBCT scans. The functions increased with beam width, tending towards equilibrium values for beam widths above about 400 mm. There were differences in $G_c(W)$ and $G_p(W)$ functions with tube potential and these were larger for the body phantoms and increased with beam width. The results at the centre of the PE body phantom that represented the ICRU/AAPM phantom did not replicate the constant relationship with tube potential reported for MSCT scanners (ICRU 2012). For the PMMA body phantom (figures 2(d)–(f)), the variations between tube potentials at the centre and periphery of the phantom for a beam of width 200 mm were in agreement with those for a CBCT scan of the same beam width obtained with a conventional CT scanner, a Somatom Definition dual source CT scanner, using MC simulations and the stationary table mode (table 2) (Li *et al* 2014).

Although $f(0)_{\mathrm{c,m}}$ and $f(0)_{\mathrm{p,m}}$ values at 140 kV were larger than those for other tube potentials as would be expected, values for $G_{\mathrm{c}}(W)$ and $G_{\mathrm{p}}(W)$ varied in different ways depending on phantom composition. Values of $G_{\mathrm{c}}(W)$ increased with tube potential for all phantoms, while those for $G_{\mathrm{p}}(W)$ decreased with tube potential for the PMMA and water phantoms, so that the differences in $G_{\mathrm{c}}(W)$ and $G_{\mathrm{p}}(W)$ tended to cancel to some extent in the derivation of $G_{\mathrm{a}}(W)$ (figures 2 and 4). The influence of tube potential on values for the $G_x(W)$ functions within the PMMA body phantom was in agreement with that found for a conventional CT scanner (table 2) (Li *et al* 2014), where $G_x(W)$ at 80 kV was lower at the centre of the phantom but larger than other tube potentials at the periphery. Whereas for the PE phantoms both $G_{\mathrm{c}}(W)$ and $G_{\mathrm{p}}(W)$ increased with tube potential (figure 3). The variations resulted from the influence of tube potential on the dose level at the centre and periphery of each phantom. Figures 3 and 4 present measurements in phantoms made from PE and water, and can be compared with the standard PMMA phantoms in figure 2. The reason for their contrasting behaviour relates to their different chemical compositions (table 1), and the resulting energy dependence of the mass energy absorption coefficients, which determine the attenuation and distribution of energy absorption within the phantoms. Polyethylene contains only carbon and hydrogen atoms, whereas water and PMMA contain oxygen atoms. Considering changes in the attenuation as photon energy is reduced (<50 keV), the attenuation starts to rise more rapidly at slightly higher photon energies in PMMA and water due to the photoelectric absorption component than it does for PE. As a result, differences in $G_{\mathrm{c}}(W)$ with tube potential are greater for PMMA (figures 2(a) and (d)) and water (figures 4(a) and (d)), than for PE (figures 3(a) and

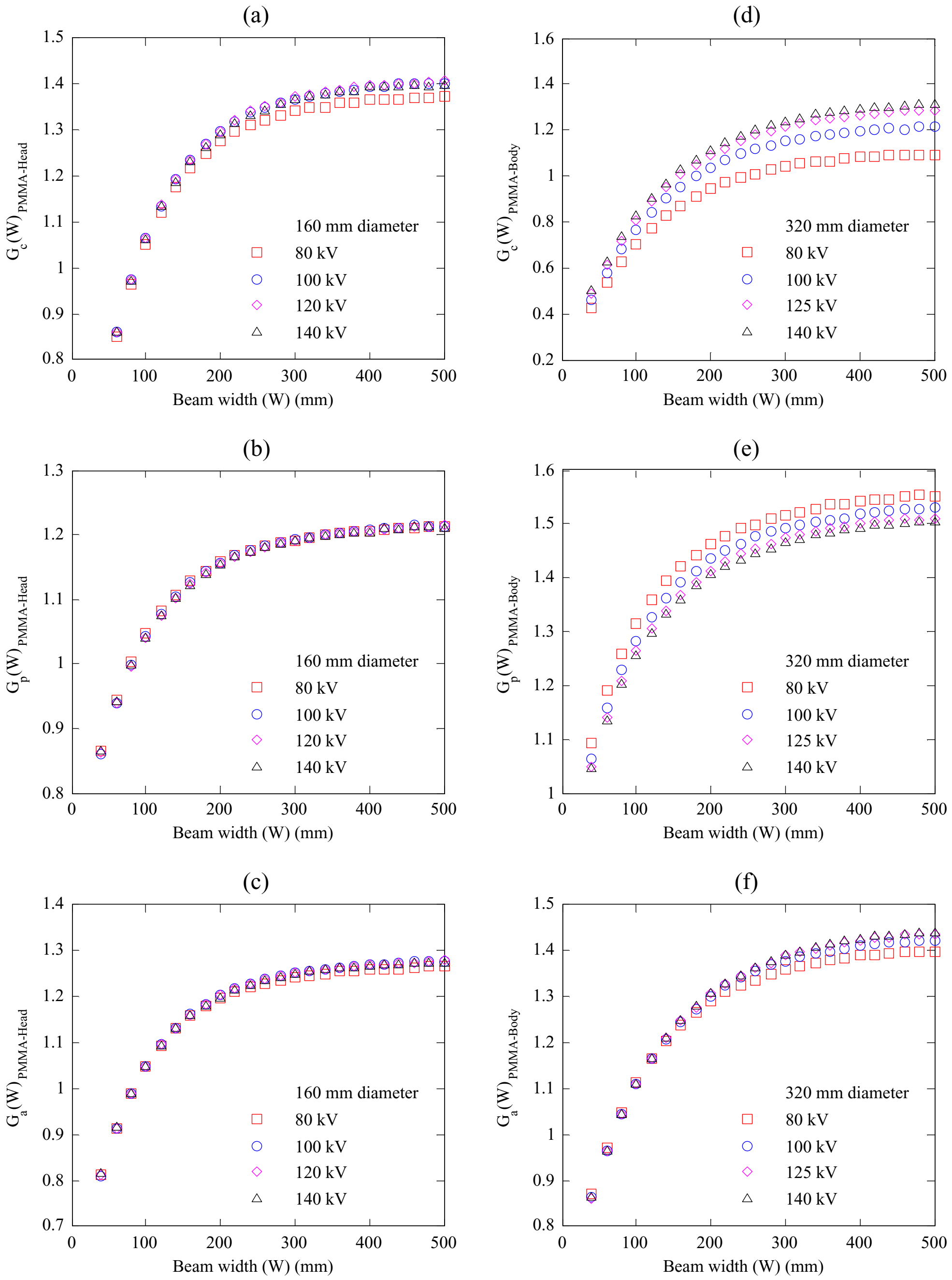

Figure 2. $G_c(W)$, $G_p(W)$, and $G_a(W)$ functions for $f(0)_{x,\ \mathrm{PMMA}}$ values calculated within 600 mm long PMMA phantoms and normalized with respect to $\mathrm{CTDI}_{w,40}$ values calculated within 150 mm long PMMA phantoms as in equation (5) using beams of width $W = 40$–500 mm. (a)–(c) for the head phantoms and (d)–(f) for the body phantoms (table 1).

Table 2. A comparison between $G_c(W)$ and $G_p(W)$ functions obtained in this study for a beam of width 200 mm calculated within the PMMA body phantoms as in equation (5) at 80–140 kV and those from Li *et al* (2014).

	This study		(Li *et al* 2014)	
Function	Centre	Periphery	Centre	Periphery
$G_x(W = 200)$–80 kV	0.94	1.46	0.88	1.78
$G_x(W = 200)$–100 kV	1.04	1.44	0.94	1.75
$G_x(W = 200)$–120/125 kV[a]	1.09	1.41	0.97	1.74
$G_x(W = 200)$–140 kV	1.11	1.40	0.99	1.73

[a] 125 kV was used for the body phantoms in this study, whereas 120 kV was used in (Li *et al* 2014).

(d), while $G_p(W)$ values are greater for lower tube potentials for PE (figures 3(b) and (e)) than for PMMA and water, because the backscatter component is larger and the attenuation lower.

The beam qualities for the kV system used in this study are similar to those for conventional CT scanners, the half value layers (HVLs) at 125 kV and 100 kV being 8.7 mm and 7.55 mm Al, respectively. Moreover, the variations with tube potential found in this study are similar in form to those reported for a CBCT scan obtained with a conventional CT scanner (table 2), although slightly larger in magnitude. Therefore, it is likely that the variations found between the $G_x(W)$ functions (figures (2)–(4)) are not strongly influenced by the type of scanner, CT or CBCT, but rather by use of the wider beams employed for CBCT scans. The main difference between the $G_x(W)$ equation (4) and $G_x(L)$ equation (2) functions is the use of beam width (W) instead of scan length (L). In a conventional CT scanner, $D_L(0)_x$ measurements at the centre and periphery of a phantom are built up from rotations of the same fan beam (≤40 mm) covering the specified length, and subsequently $D_L(0)_x$ values are normalized with respect to CTDI_{vol} for the same beam width. However, in a CBCT scan, the beam is varied over a wide range of widths to measure $f(0)_{x,\text{m}}$, and subsequently normalized with respect to CTDI_w for a reference beam. The influence of scanning parameters at the centre and periphery of the phantom varies accordingly to the change in geometry as the beam width increases from 40 to 500 mm. Photons near the edges of wider beams pass through the phantoms at oblique angles, and are therefore more heavily attenuated. This will change the relative magnitudes of doses measured at the centres and peripheries of the phantoms, as well as the path length through the phantoms contributing to the scatter. All of these factors will change with tube potential, which alters the photon energy distribution within the beams. Moreover, these differences will also vary with the attenuation and scattering properties of the phantom material. For a conventional CT scanner, the variation in beam geometry along the phantom is comparatively small, and normalization of $G_x(L)$ with respect to the CTDI_{vol}, which averages the dose over the x–y plane, is capable of eliminating most of the scan parameter differences. This view is supported by the lower differences found at $W = 40$ mm compared to those for wider beams, which increased with beam width (figures (2)–(4)(a) and (d)). However, the CTDI_w for the narrow reference beam that has been used for normalization of $G_x(W)$ equation (4) will not give the same averaging of the dose that occurs with the different widths of cone beams. Therefore, the differences in geometry of the beams is considered to be the main reason for the discrepancy found between the results at different tube potentials obtained in this study at the centre of the PE body phantom (figure 3(d)) and those reported at the centre of the ICRU/AAPM phantom for a conventional CT scanner (ICRU 2012).

The discrepancies between values of the proposed function $G_x(W)$ equation (4) for different tube potentials demonstrate an inherent weakness in its application for CBCT scans. These

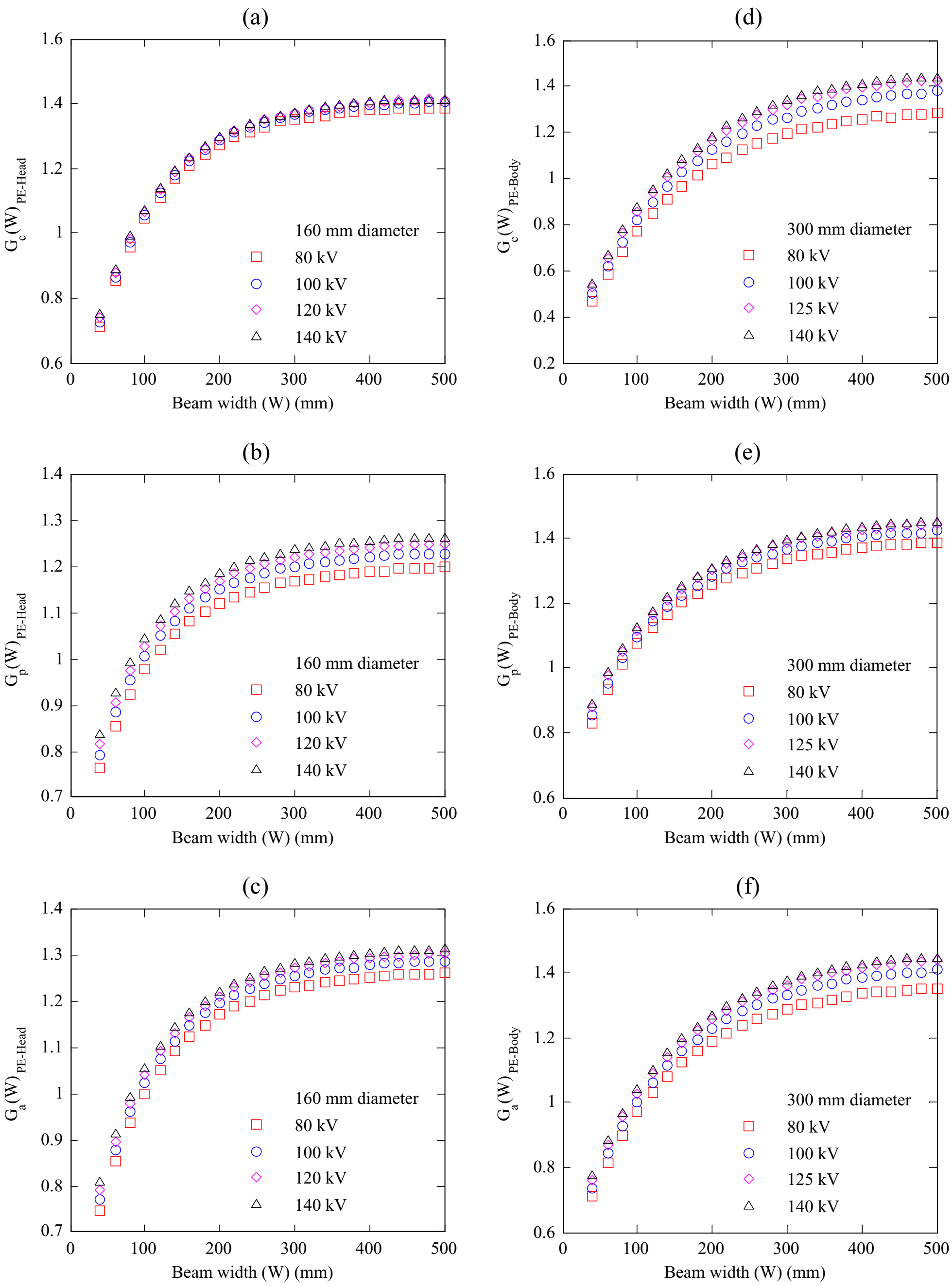

Figure 3. $G_c(W)$, $G_p(W)$, and $G_a(W)$ functions for $f(0)_{x,\mathrm{PE}}$ values calculated within 600 mm long PE phantoms and normalized with respect to $\mathrm{CTDI}_{w,40}$ values calculated within 150 mm long PMMA phantoms as in equation (5) using beams of width W = 40–500 mm. (a)–(c) for the head phantoms and (d)–(f) for the body phantoms (table 1).

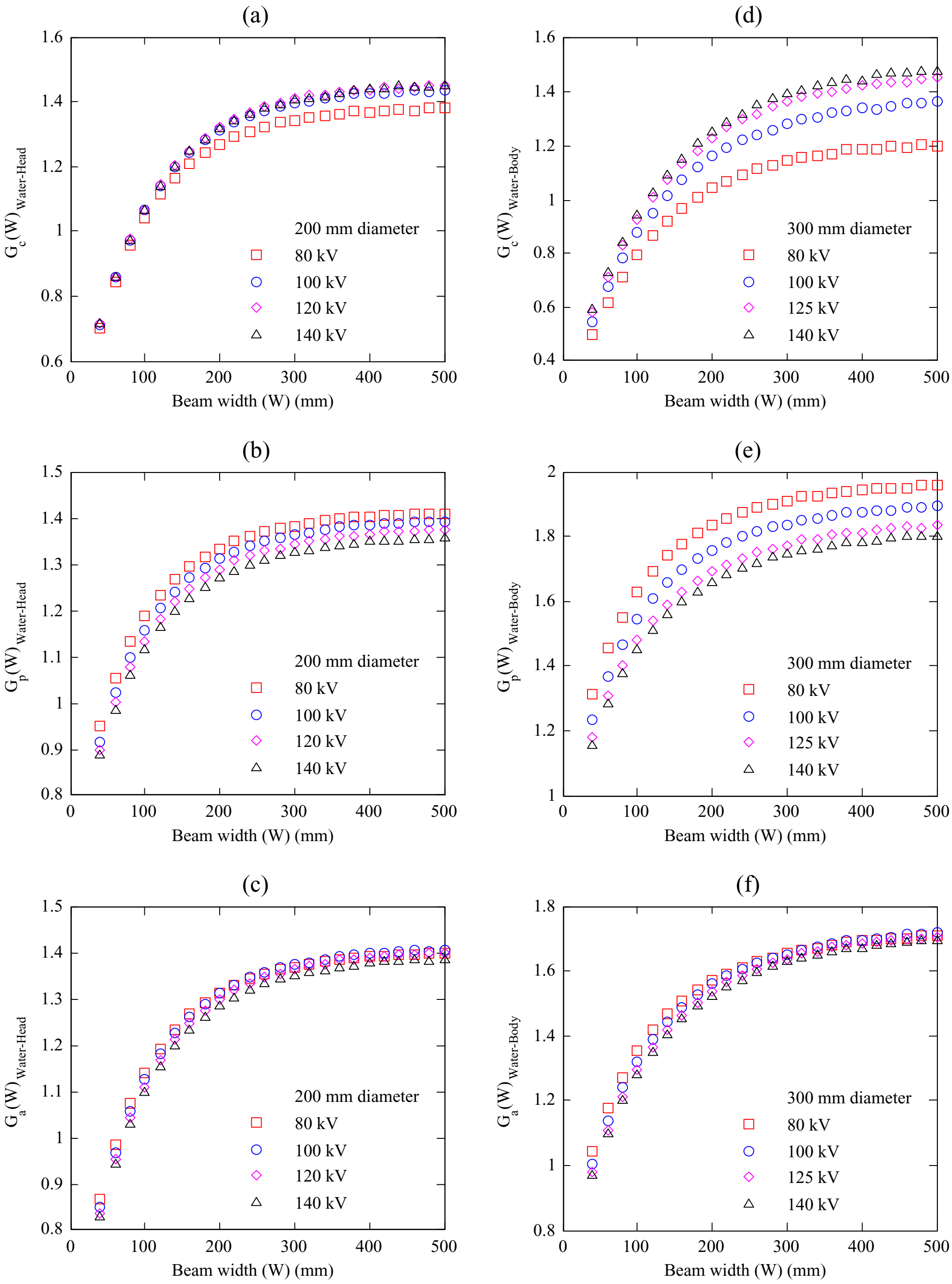

Figure 4. $G_c(W)$, $G_p(W)$, and $G_a(W)$ functions for $f(0)_{x,\mathrm{Water}}$ values calculated within 600 mm long water phantoms and normalized with respect to $\mathrm{CTDI}_{w,40}$ values calculated within 150 mm long PMMA phantoms as in equation (5) using beams of width W = 40–500 mm. (a)–(c) for the head phantoms and (d)–(f) for the body phantoms (table 1).

could be reduced by normalizing with respect to variables that take more account of the differences in scanning parameters and beam width such as:

(1) To normalize $f(0)_{x,\mathrm{m}}$ with respect to $\mathrm{CTDI}_{100,x}$ measured at the same position within the phantom (centre or periphery) instead of CTDI_w
(2) To normalize every value of $f(0)_{x,\mathrm{m}}$ with respect to $\mathrm{CTDI}_{100,x}$ measured using the same beam width (W) used for that $f(0)_{x,\mathrm{m}}$ measurement instead of a reference beam width.

For the first factor, figure 5 shows $f(0)_{x,\mathrm{m}}$ values calculated within body phantoms of different compositions for beams of width 40–500 mm normalized with respect to $\mathrm{CTDI}_{100,x}$ obtained at the same position within the standard PMMA body phantom using a reference beam width of 40 mm. This factor reduced the variations with tube potential significantly at some positions, such as at the centre of the PMMA body phantom, but did not reduce the differences in others, such as for the periphery of the PE phantom. Results for the head phantoms were similar to those for the body phantoms (figure 5), but the variations were lower. However, when the second factor was combined with the first, taking account of the different beam geometries, the variations with tube potential declined remarkably at all the positions for both the body phantoms (figure 6) and head phantoms. The application of the two factors appeared not only to cancel out the influence of the scanning parameters, but also the type of the scanner, since results for beams of width 40–240 mm calculated within the PMMA phantoms were in good agreement with those for a conventional CT scanner (Li *et al* 2014) (figures 6(a)–(b), and the differences shown in table 2 were eliminated. As the range of beam widths used in the present study was wider than that in (Li *et al* 2014) (30–250) mm, a discrepancy between the results outside this range was anticipated as clearly seen in figure 6(a). The curves for $f(0)_{x,\mathrm{m}}/\mathrm{CTDI}_{100,x}$ (figure 6) were entirely different from those for the $G_x(W)$ function (figures (2)–(4)), as expected from use of the same beam width to measure $f(0)_{x,\mathrm{m}}$ and $\mathrm{CTDI}_{100,x}$ as discussed in section 2.2.

Although the normalisation of $f(0)_{x,\mathrm{m}}$ with respect to $\mathrm{CTDI}_{100,x}$ measured at the same position and beam width (figure 6) allowed the variations in the $G_x(W)$ functions with tube potential to be minimized (figures (2)–(4)), the use of $\mathrm{CTDI}_{100,x}$ for beams of width >100 mm does not provide a good practical solution as CTDI_{100} values decline significantly when the primary beam width becomes wider than 100 mm (Boone 2007, Kyriakou *et al* 2008, Abuhaimed *et al* 2014). The influence of this decline on the $f(0)_{x,\mathrm{m}}/\mathrm{CTDI}_{100,x}$ values can be seen clearly in figure 6 as the function starts to rise more rapidly at $W > 100$ mm. Therefore, a modification to the denominator of the $G_x(W)$ function equation (4) is proposed to achieve an equation that will tend to an asymptotic function for wide beams, in a similar manner to the $G_x(L)$ function, but still be independent of scan parameters.

3.2. Investigation of a modified function $G_x(W)_{100}$ for CBCT scan dosimetry

The main concept of the $G_x(L)$ and $G_x(W)$ functions is to evaluate the cumulative doses $D_L(0)_x$ and $f(0)_x$, while preserving the use of the CT dosimetry system, with the 100 mm pencil ionization chamber and the standard PMMA phantoms, which are available worldwide. Therefore, a modification should be applied to the denominator of the $G_x(W)$ function, $\mathrm{CTDI}_{\mathrm{w,ref}}$. taking into account the influence of the two factors discussed in section 3.1. A practical approach to CBCT dosimetry has been proposed by (Amer *et al* 2007), based on evaluation of the dose in the middle of a scan of the standard PMMA phantoms averaged over the 100 mm length of a pencil ionization chamber. This dosimetry parameter has been studied using different CT or CBCT scanners (Geleijns *et al* 2009, Li *et al* 2014, Abuhaimed *et al* 2015). It is defined as:

$$f_{100}(150)_x = \frac{1}{100}\int_{-50\ \mathrm{mm}}^{50\ \mathrm{mm}} D(z)_x\, \mathrm{d}z \quad (6)$$

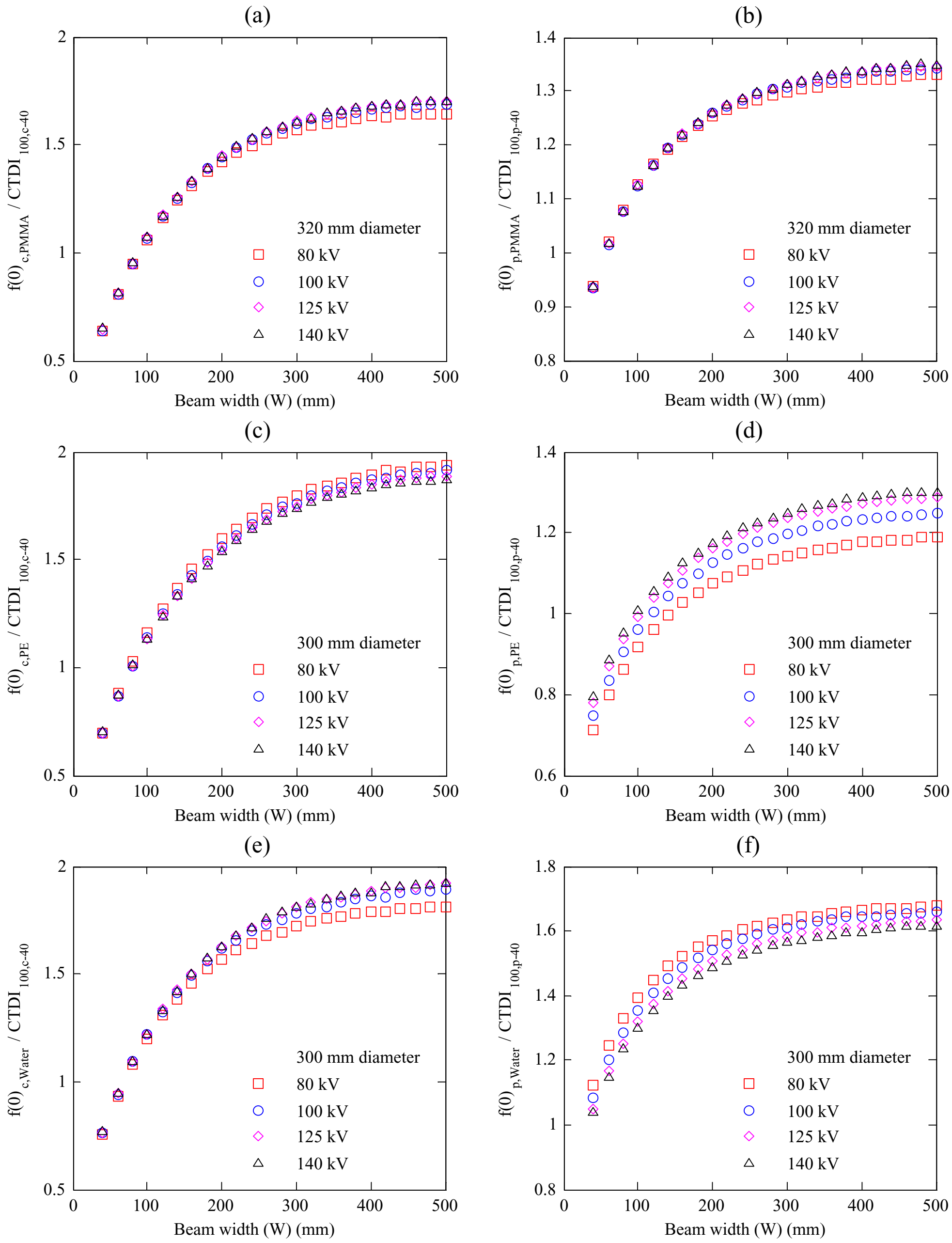

Figure 5. $f(0)_{x,\mathrm{m}}$ values calculated within 600 mm long body phantoms of different compositions using beams of width W = 40–500 mm and normalized with respect to $\mathrm{CTDI}_{100,x\text{–}40}$ measured at the same position within a 150 mm long PMMA body phantom (centre and periphery) using a reference beam of width $W = 40$ mm. (a)–(b) PMMA, (c)–(d) PE, and (e)–(f) water body phantoms (table 1).

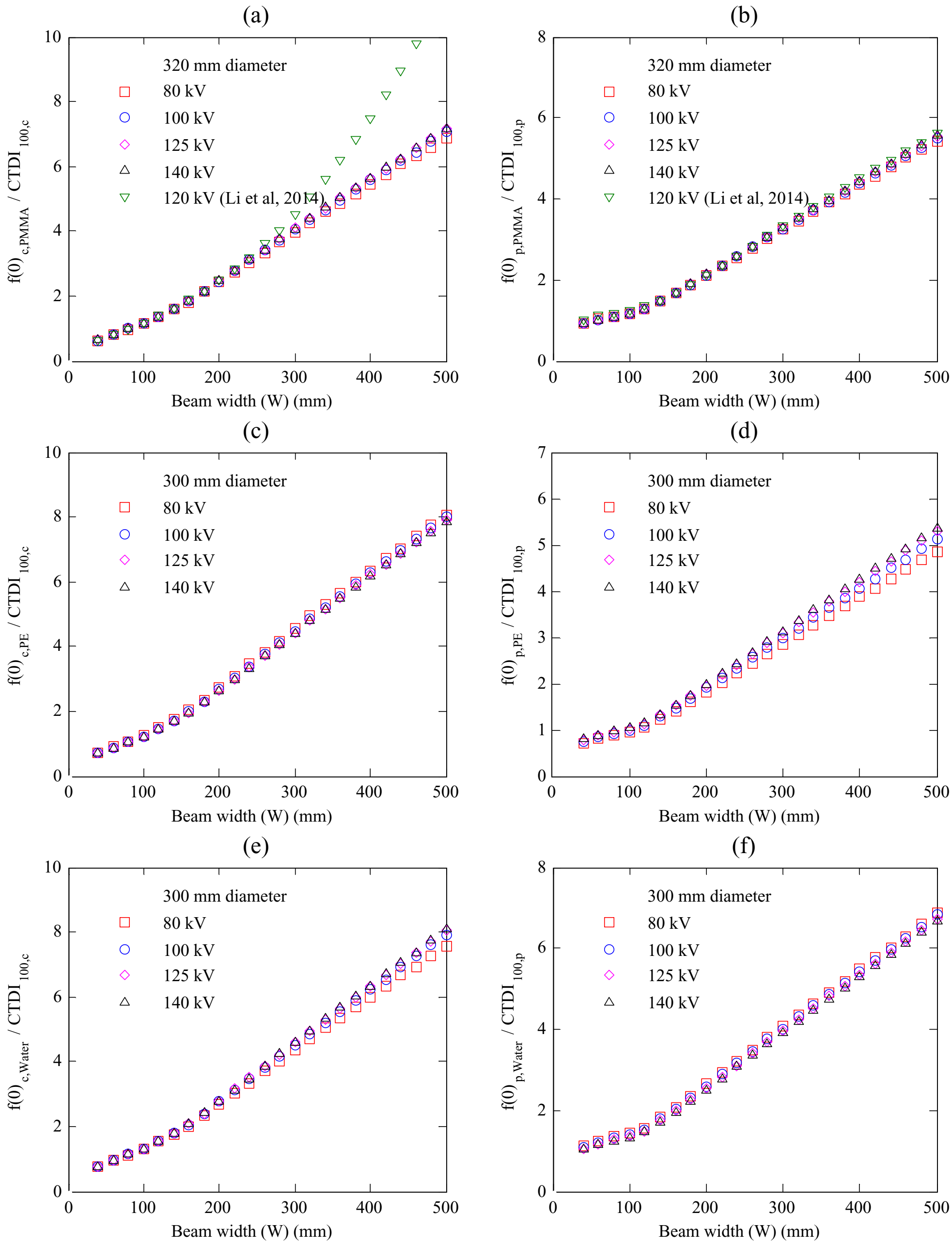

Figure 6. $f(0)_{x,\mathrm{m}}$ values calculated within 600 mm long body phantoms of different compositions using beams of width W = 40–500 mm and normalized with respect to $\mathrm{CTDI}_{100,x}$ measured at the same position within a 150 mm long PMMA body phantom (centre and periphery) and with the same beam widths. (a)–(b) PMMA, (c)–(d) PE, and (e)–(f) water body phantoms (table 1). PMMA results are compared with values reported in Li *et al* (2014) at 120 kV.

where 100 and 150 indicate lengths of the ionization chamber and the standard PMMA phantoms, respectively, and x is the central or peripheral position within the phantom. This dosimetry quantity will take more account of the contributions of scatter from wider beams within the central region of the phantom. Thus, it is proposed for use in the denominator of a modified function defined as:

$$G_x(W)_{100} = \frac{Nf(0)_x}{f_{100}(150)_x} \tag{7}$$

As CTDI_{100} and $f_{100}(150)_x$ values are measured in the same manner for equations (A.1) and (6), the MC results discussed in section 3.1 were re-analyzed to calculate $G_x(W)_{100}$ as:

$$G_\mathrm{c}(W)_{100} = \frac{f(0)_\mathrm{c,m}}{f_{100}(150)_\mathrm{c}} \quad G_\mathrm{p}(W)_{100} = \frac{f(0)_\mathrm{p,m}}{f_{100}(150)_\mathrm{p}} \quad G_\mathrm{a}(W)_{100} = \frac{f(0)_\mathrm{w,m}}{f_{100}(150)_\mathrm{w}} \tag{8}$$

Figures (7)–(9) show the $G_x(W)_{100}$ functions calculated as described in equation (8) for the phantoms studied (table 1). The functions show a weak dependency on tube potential with variations at some positions such as the periphery of the PE phantoms. However, the variations were much smaller than those found for the $G_x(W)$ functions (figures (2)–(4)), and did not vary significantly with beam width. The curves for the $G_x(W)_{100}$ functions were different from those for $G_x(L)$ and $G_x(W)$, because of the form of $f_{100}(150)_x$ in the denominator. $G_x(W)_{100}$ decreased steadily with beam width until ~150 mm, after which the functions rose slightly. For beams of width <150 mm, $f(0)_{x,\mathrm{m}}$ was greater than $f_{100}(150)_x$, but differences between the values decreased with increasing beam width. Values of $f_{100}(150)_x$ were virtually constant for beams wider than 150 mm, as the outer parts of the beams did not interact with the phantom, whereas $f(0)_{x,\mathrm{m}}$ values continued to rise approaching the equilibrium value $f_\mathrm{eq}(0)_x$. This explains the slight increase in $G_x(W)_{100}$ for beams wider than 150 mm, especially at the centres of the phantoms. The relationship between the values for $f_{100}(150)_x$ and $f(0)_x$ have been discussed in detail in (Abuhaimed *et al* 2015).

The $G_x(W)_{100}$ function described in equation (7) provides an option for CBCT scans, which is relatively independent of tube potential in the range 80–140 kV (figures (7)–(9)), that could be used for evaluating cumulative doses in long phantoms. This approach is well suited to CBCT applications such as those in radiotherapy, where the number of beam diameters and scanning parameters used is limited for the majority of examinations, and measurements of the $f_{100}(150)_x$ within standard dosimetry phantoms can readily be made for the beam widths used clinically. The $G_x(W)_{100}$ functions calculated within the PMMA phantoms were in good agreement with those calculated in (Li *et al* 2014) within the range 40–240 mm (figure 7). The discrepancy between the results outside this range was anticipated as explained earlier.

3.3. Derivation of values of $G_x(W)_{100}$ for use in CBCT dosimetry

The $G_x(W)_{100}$ function shown in figures (7)–(9) were fitted to sixth-order polynomial equations. Coefficients of the fitted equations are given in table 3, and these can be used to evaluate $f(0)_{x,\mathrm{m}}$ for any beam width in the range 40–500 mm at tube potentials 80–140 kV within the PMMA, PE, or water head and body phantoms used in this study (table 1). All that is required are measurements of $f_{100}(150)_x$ within the standard PMMA phantoms for the beam width (W) being used. Subsequently, $f(0)_{x,\mathrm{m}}$ is calculated as follows:

$$f(0)_{x,\mathrm{m}} = G_x(W)_{100} \times f_{100}(150)_x \tag{9}$$

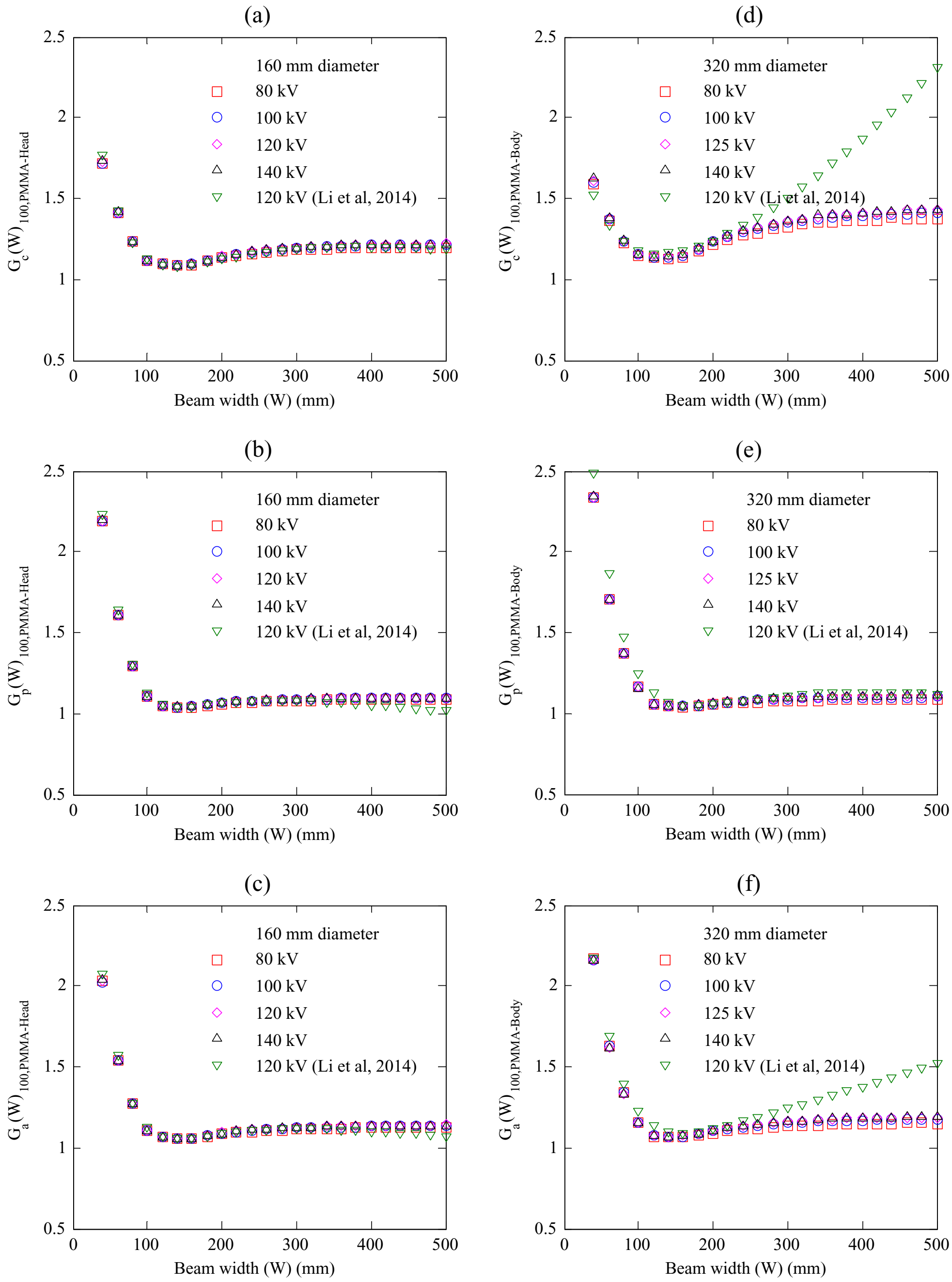

Figure 7. $G_c(W)_{100}$, $G_p(W)_{100}$, and $G_a(W)_{100}$ functions for $f(0)_{x,\mathrm{PMMA}}$ values calculated within the 600 mm long PMMA phantoms for beams of width 40–500 mm and normalized with respect to $f_{100}(150)_x$ values calculated within the standard 150 mm long PMMA phantoms as in equation (8). (a)–(c) for the head phantoms and (d)–(f) for the body phantoms (table 1). Results are compared with values reported in Li *et al* (2014) at 120 kV.

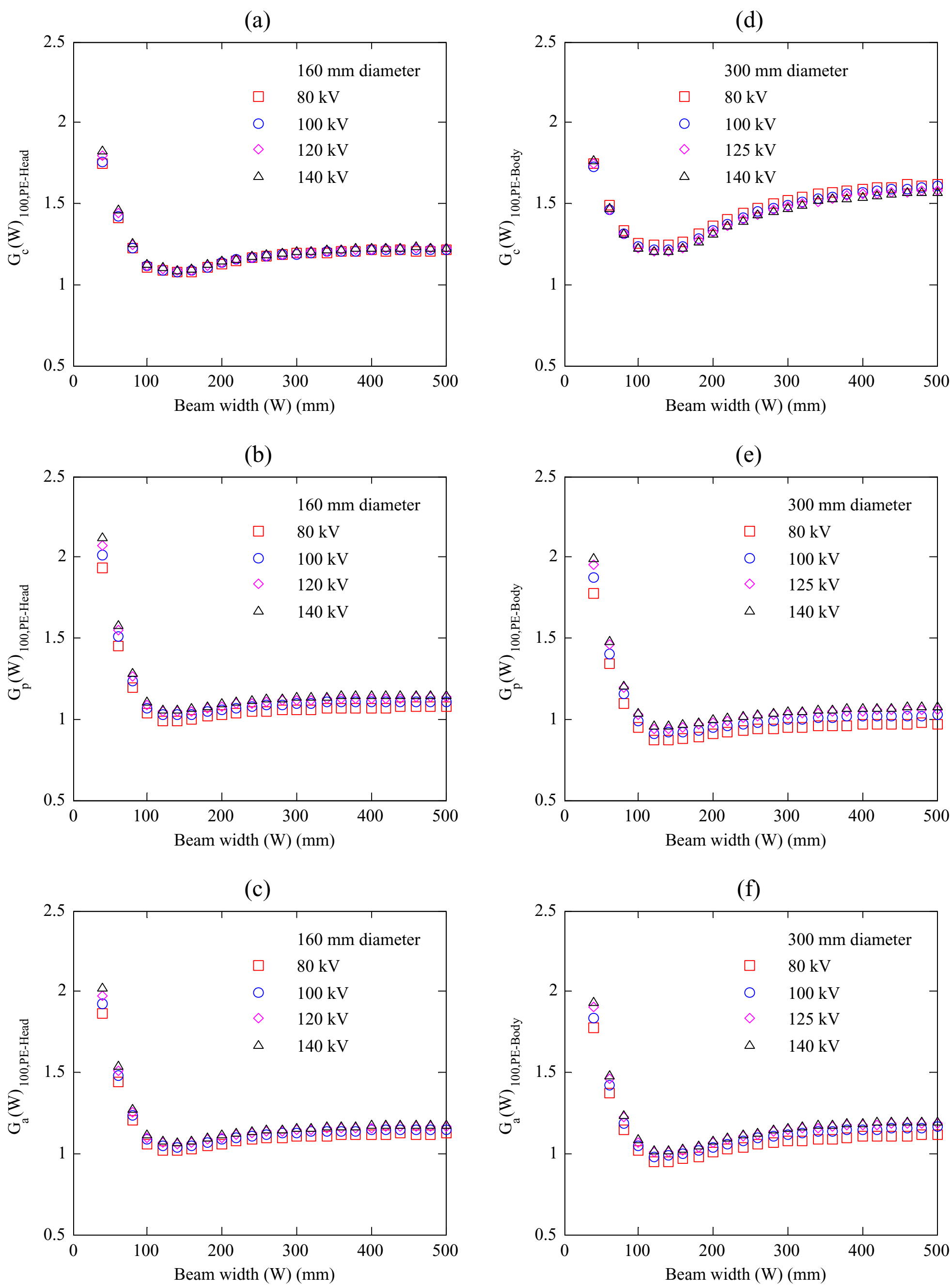

Figure 8. $G_c(W)_{100}$, $G_p(W)_{100}$, and $G_a(W)_{100}$ functions for $f(0)_{x,\mathrm{PE}}$ values calculated within the 600 mm long PE phantoms for beams of width 40–500 mm and normalized with respect to $f_{100}(150)_x$ values calculated within the standard 150 mm long PMMA phantoms as in equations (8). (a)–(c) for the head phantoms and (d)–(f) for the body phantoms (table 1).

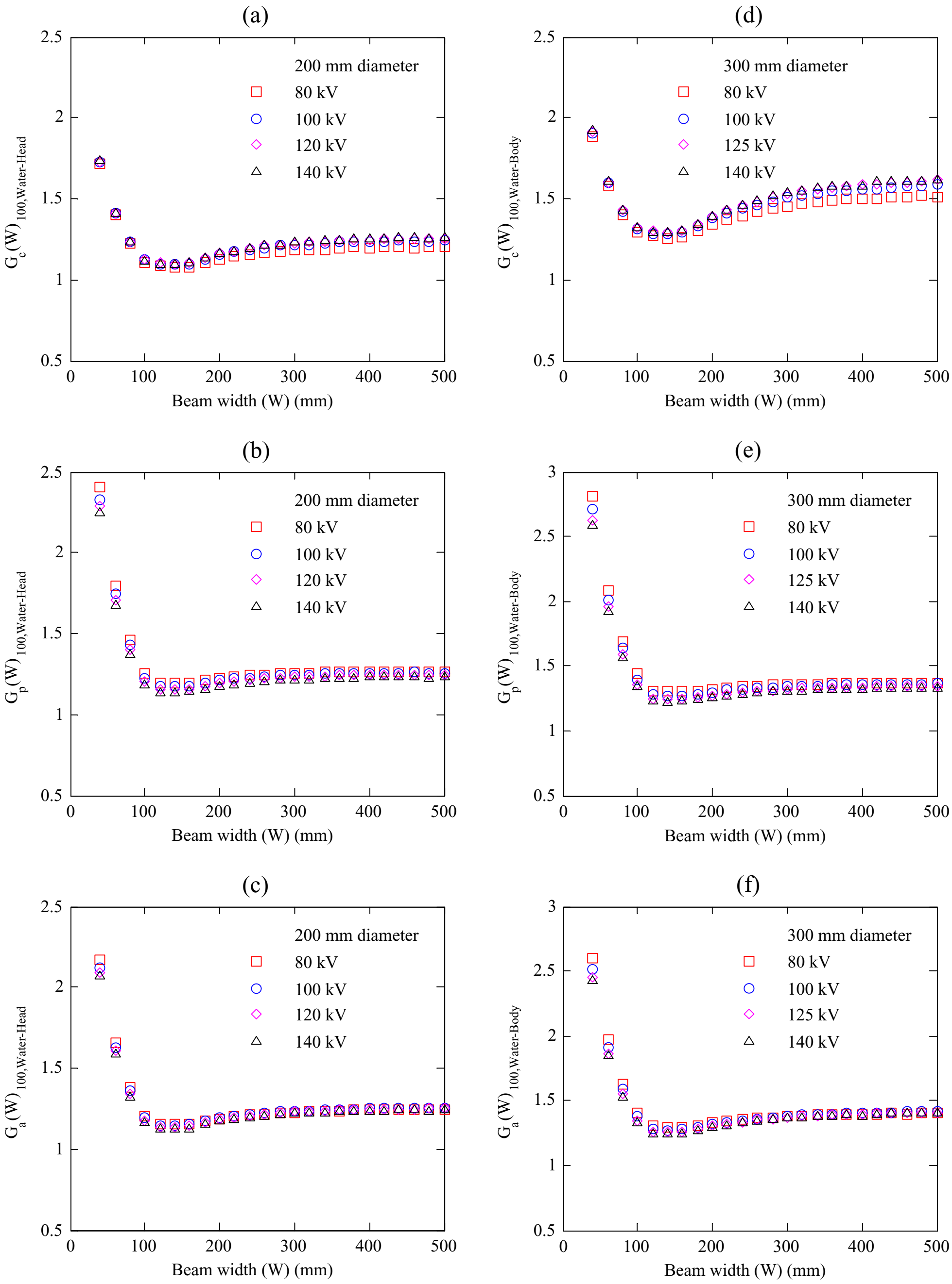

Figure 9. $G_c(W)_{100}$, $G_p(W)_{100}$, and $G_a(W)_{100}$ functions for $f(0)_{x,\mathrm{Water}}$ values calculated within the 600 mm long water phantoms for beams of width 40–500 mm and normalized with respect to $f_{100}(150)_x$ values calculated within the standard 150 mm long PMMA phantoms as in equation (8). (a)–(c) for the head phantoms and (d)–(f) for the body phantoms (table 1).

Table 3. Coefficients and R^2 of equations for $G_x(W)_{100}$ as a function of beam width W (in mm) for relationships shown in figures (7)–(9). The functions were fitted to sixth-order polynomial equations for beams of width 40–500 mm at tube potentials 80–140 kV.

$$G_x(W)_{100} = P_1 W^6 + P_2 W^5 + P_3 W^4 + P_4 W^3 + P_5 W^2 + P_6 W + P_7$$

	Head			Body		
Coefficients	Centre	Periphery	Weighted	Centre	Periphery	Weighted
PMMA						
p_1	2.721×10^{-15}	6.906×10^{-15}	5.459×10^{-15}	9.872×10^{-16}	6.396×10^{-15}	5.057×10^{-15}
p_2	-5.402×10^{-12}	-1.304×10^{-11}	-1.040×10^{-11}	-2.301×10^{-12}	-1.223×10^{-11}	-9.778×10^{-12}
p_3	4.339×10^{-09}	9.891×10^{-09}	7.975×10^{-09}	2.154×10^{-09}	9.429×10^{-09}	7.636×10^{-09}
p_4	-1.794×10^{-06}	-3.841×10^{-06}	-3.135×10^{-06}	-1.028×10^{-06}	-3.736×10^{-06}	-3.070×10^{-06}
p_5	3.982×10^{-04}	8.001×10^{-04}	6.618×10^{-04}	2.585×10^{-04}	7.980×10^{-04}	6.659×10^{-04}
p_6	-4.388×10^{-02}	-8.393×10^{-02}	-7.017×10^{-02}	-3.076×10^{-02}	-8.640×10^{-02}	-7.283×10^{-02}
p_7	2.947	4.484	3.956	2.483	4.728	4.183
R^2	0.99	0.99	0.99	0.98	0.99	0.99
PE						
p_1	3.243×10^{-15}	5.916×10^{-15}	4.996×10^{-15}	1.360×10^{-15}	4.552×10^{-15}	3.768×10^{-15}
p_2	-6.347×10^{-12}	-1.124×10^{-11}	-9.556×10^{-12}	-3.016×10^{-12}	-8.814×10^{-12}	-7.400×10^{-12}
p_3	5.022×10^{-09}	8.587×10^{-09}	7.362×10^{-09}	2.711×10^{-09}	6.890×10^{-09}	5.879×10^{-09}
p_4	-2.046×10^{-06}	-3.360×10^{-06}	-2.909×10^{-06}	-1.255×10^{-06}	-2.772×10^{-06}	-2.407×10^{-06}
p_5	4.481×10^{-04}	7.050×10^{-04}	6.170×10^{-04}	3.090×10^{-04}	6.010×10^{-04}	5.314×10^{-04}
p_6	-4.885×10^{-02}	-7.417×10^{-02}	-6.551×10^{-02}	-3.611×10^{-02}	-6.562×10^{-02}	-5.861×10^{-02}
p_7	3.135	4.065	3.746	2.769	3.719	3.492
R^2	0.99	0.98	0.98	0.99	0.96	0.96
Water						
p_1	2.830×10^{-15}	6.777×10^{-15}	5.397×10^{-15}	1.707×10^{-15}	6.711×10^{-15}	5.471×10^{-15}
p_2	-5.607×10^{-12}	-1.286×10^{-11}	-1.032×10^{-11}	-3.652×10^{-12}	-1.292×10^{-11}	-1.063×10^{-11}
p_3	4.493×10^{-09}	9.813×10^{-09}	7.955×10^{-09}	3.183×10^{-09}	1.004×10^{-08}	8.347×10^{-09}
p_4	-1.853×10^{-06}	-3.836×10^{-06}	-3.144×10^{-06}	-1.434×10^{-06}	-4.012×10^{-06}	-3.378×10^{-06}
p_5	4.102×10^{-04}	8.043×10^{-04}	6.671×10^{-04}	3.462×10^{-04}	8.649×10^{-04}	7.378×10^{-04}
p_6	-4.493×10^{-02}	-8.469×10^{-02}	-7.089×10^{-02}	-4.045×10^{-02}	-9.434×10^{-02}	-8.119×10^{-02}
p_7	2.975	4.637	4.064	3.051	5.301	4.756
R^2	0.98	0.99	0.99	0.96	0.99	0.99

Table 4. Experimental measurements of $f(0)_{x,\mathrm{PMMA}}$ within PMMA head and body phantoms 450 mm in length and $f_{100}(150)_x$ in standard 150 mm long PMMA phantoms, using the clinical beam width 198 mm at 100 kV for the head and 125 kV for the body phantom. The $G_x(W)_{100}$ functions for PMMA were calculated from the experimental measurements (Exp) using equation (8) and from the equations (table 3) derived from MC results. The $G_x(W)_{100}$ functions for PE and water were calculated from the equations (table 3) as in equation (9). [Difference (%) = ($G_x(W)_{100,\mathrm{PMMA\text{-}MC}}$ – $G_x(W)_{100,\mathrm{PMMA\text{-}Exp}}$ / $G_x(W)_{100,\mathrm{PMMA\text{-}Exp}}$) × 100].

	Head phantom			Body phantom		
	Centre	Periphery	Weighted	Centre	Periphery	Weighted
			PMMA			
$f(0)_{x,\mathrm{PMMA\text{-}Exp}}$ mGy/100 mAs	2.50	2.26	2.34	1.62	1.98	1.86
$f_{100}(0,150)_x$ mGy/100 mAs	2.32	2.18	2.22	1.29	1.84	1.66
$G_x(W = 198)_{100,\mathrm{PMMA\text{-}Exp}}$	1.08	1.04	1.06	1.26	1.08	1.12
$G_x(W = 198)_{100,\mathrm{PMMA\text{-}MC}}$	1.13	1.07	1.09	1.22	1.06	1.10
Difference (%)	4.63	2.88	2.83	– 3.17	– 1.85	– 1.79
			PE			
$G_x(\mathrm{W} = 198)_{100,\mathrm{PE\text{-}MC}}$	1.13	1.07	1.09	1.33	0.96	1.04
$f(0)_{x,\mathrm{PE}}$ mGy/100 mAs	2.63	2.33	2.42	1.71	1.76	1.74
			Water			
$G_x(\mathrm{W} = 198)_{100,\mathrm{Water\text{-}MC}}$	1.15	1.20	1.18	1.37	1.29	1.31
$f(0)_{x,\mathrm{Water}}$ mGy/100 mAs	2.66	2.62	2.62	1.76	2.37	2.18

Table 4 shows experimental measurements for $f(0)_{x,\mathrm{PMMA}}$ made within PMMA head and body phantoms of length 450 mm, and for $f_{100}(150)_x$ measured within the standard 150 mm long PMMA phantoms using the clinical beam width 198 mm. These measurements were used to evaluate the $G_x(W)_{100}$ function experimentally as in equation (8). The experimental values for the $G_x(W)_{100}$ function were then compared against those calculated by MC (table 3). As shown in table 4, the experimental and MC values for the $G_x(W)_{100}$ function were in good agreement, where the average differences within the head and body phantoms were 3.45% and 2.27%, respectively.

The experimental measurements for $f(0)_{x,\mathrm{PMMA}}$ were also used to assess $f(0)_{x,\mathrm{m}}$ within the PE and water head and body phantoms using equation (9). Table 4 shows that the $f(0)_\mathrm{c}$ value within the water head phantom was slightly larger than those in the PMMA and PE head phantoms, but all were within a factor of 1.06. $f(0)_\mathrm{c}$ within the water and PE body phantoms were comparable within a factor of 1.02 and slightly larger than that for the PMMA body phantom. These findings were consistent with $\mathrm{CTDI}_{\infty,\mathrm{c}}$ values of (Zhou and Boone 2008). For the periphery, however, values for $f(0)_\mathrm{p}$ calculated within the water head and body phantoms were greater than those for the PMMA and PE phantoms. $f(0)_\mathrm{p}$ for the water head and body phantoms were 16% and 20% larger, respectively, than those in the PMMA phantoms. These findings were also in agreement with those of (Zhou and Boone 2008).

3.4. Capability of the $G_x(W)_{100}$ functions for different CT and CBCT scanners

As mentioned previously, ICRU compared the $G_c(L)_{PE}$ functions for three different conventional CT scanners and reported that the parameters of any scan are cancelled out by normalization to the $CTDI_{vol}$, and the effect of CT scanner model is also eliminated (ICRU 2012). Li *et al* (2014) showed that their results obtained with a full 360° scan using a Somatom Definition dual source CT were in good agreement with those of a C-arm flat detector CT (Siemens Axiom Artis) acquired with a partial 200° scan without bowtie filters (Kyriakou *et al* 2008), a Varian On-Board Imager (Osei *et al* 2009), and a Toshiba Aquilion ONE CT scanner (Geleijns *et al* 2009). These investigations, including the (Li *et al* 2014) study as shown in figure 7, were in good agreement with the results reported in a previous study (Abuhaimed *et al* 2015) using the same kV system involved in this investigation. The results for CBCT scans acquired with a partial 200° scan were also found to be comparable to the those acquired with a full 360° scan (Abuhaimed *et al* 2014, Abuhaimed *et al* 2015). Moreover, the efficiency values for $CTDI_{100}$ are comparable for a wide range of CT and CBCT scanners (Dixon and Ballard 2007, Li *et al* 2011, Martin *et al* 2011, Li *et al* 2012, Abuhaimed *et al* 2014). Dixon and Boone (2010) compared the approach to equilibrium functions for two MSCT scanners (GE 16 channel and 64 channel scanners) and a CBCT scanner (256-slice CT scanner) employed with the stationary table mode, and again reported good agreement in results between scanners.

These findings indicate that the effect of CT scanner is largely cancelled out when the effects of the scanning parameters are eliminated by using relative measurements, i.e. a dose ratio, of two dose indices measured with identical scanning parameters as discussed in sections 3.1 and 3.2. This may give a good indication for the suitability of utilizing the equations (table 3) for different CT or CBCT scanners employed with wide beams 40–500 mm and the stationary table mode at 80–140 kV taking into account the dependence of the functions on diameter and composition of the phantom. The functions $G_x(L)$ and $G_x(W)_{100}$ have been derived to allow the new dosimetry quantities to be calculated from measurements made with the standard equipment. The main strength of the $G_x(L)$ function proposed by (ICRU 2012) and the modified function $G_x(W)_{100}$ investigated in the present study is that they are relatively independent of scanning parameters including the tube potential. The functions enable the evaluation of $D_L(0)_x$ or $f(0)_{x,m}$ without the need for long phantoms of PMMA, PE, or water, which are heavy and difficult to handle in the scanning room and to transport between hospitals.

4. Conclusion

The aim of this study was to extend application of the $G_x(L)$ function, which can be used to derive the cumulative dose $D(0)_L$ in larger phantoms for MSCT scans (ICRU 2012), to CBCT scans with stationary tables. The main difference is that the beam width (W) determines the length of the scan along the rotation axis (z–axis) rather than the number and pitch of the tube rotations. An analogous function $G_x(W)$ normalized with respect to $CTDI_w$ of a narrow reference beam showed a dependence on tube potential that varied with phantom composition. This differed from results at the centre of the ICRU/AAPM PE body phantom reported for conventional CT scanners (ICRU 2012). This is considered to be due to differences in geometry between the narrow fan beams used in MSCT scans and wide beams used for CBCT scans. A modified function $G_x(W)_{100}$, for which cumulative doses $f(0)_{x,m}$ are normalized with respect to an alternative dose index $f_{100}(150)_x$ measured by a 100 mm ionization chamber with similar scan parameters and beam width at the same position within standard PMMA phantoms, was found to be relatively independent of beam quality over the range of tube potentials 80–140 kV. The $G_x(W)_{100}$ functions have been fitted to sixth-order polynomial equations, which can be utilized to evaluate $f(0)_{x,\mathrm{m}}$ within infinitely long PMMA, PE and water head and body phantoms. These are valid for beam widths of 40–500 mm at 80–140 kV at the centres and peripheries of the phantoms. They allow cumulative doses $f(0)_{x,\mathrm{m}}$ in long phantoms to be evaluated from measurements of $f_{100}(150)_x$ at the centre and periphery of the standard PMMA phantoms for the beam of interest. It may be possible to apply the fitted equations to calculate cumulative doses for any CT or CBCT scan acquired with stationary table mode.

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Appendix A

CTDI_{100} is the base for other CT dosimetry quantities. CTDI_{100} is defined as:

$$\mathrm{CTDI}_{100} = \frac{1}{N \times T} \int_{-50\ \mathrm{mm}}^{50\ \mathrm{mm}} D(z)\,\mathrm{d}z \tag{A.1}$$

where $D(z)$ is the dose profile resulting from a single axial scan, N is the number of slices acquired in a single scan, and T is the nominal thickness of a single slice. The product of $(N \times T)$ is equal to the nominal beam width involved in the scan.

The CTDI_{100} measurements are made within the standard PMMA phantoms either at the centre ($\mathrm{CTDI}_{100,\mathrm{c}}$) or at four peripheral positions situated 10 mm below the phantom surface ($\mathrm{CTDI}_{100,\mathrm{p}}$). The weighted CTDI_{100} known as CTDI_w accounts for inhomogeneity of the dose distribution over the axial scan plane, and is defined as:

$$\mathrm{CTDI}_\mathrm{w} = \frac{1}{3}\,\mathrm{CTDI}_{100,\mathrm{c}} + \frac{2}{3}\,\mathrm{CTDI}_{100,\mathrm{p}} \tag{A.2}$$

where $\mathrm{CTDI}_{100,\mathrm{p}}$ is the average of the four dose measurements made at the peripheral positions, ($\mathrm{CTDI}_{100,\mathrm{p}} = 1/4 \sum_{n=1}^{4} \mathrm{CTDI}_{100,\mathrm{pn}}$). For MSCT scans, ($\mathrm{CTDI}_\mathrm{vol} = \mathrm{CTDI}_\mathrm{w}$ / pitch) is used to quantify the dose along the scan axis to allow for differences in the beam width and pitch of the tube rotation, where the pitch is the ratio of the table movement per a single rotation (b) to the nominal beam width ($N \times T$), (pitch = b/$N \times T$). Moreover, DLP = $\mathrm{CTDI}_\mathrm{vol} \times$ the scan length (L) is employed in assessment of dose for patients undergoing a MSCT scan. It should be noted that CTDI_{100} (equation A.1) used in this study was the standard CTDI_{100}. This is different from the modified CTDI_{100} proposed by IEC (IEC 2010) and investigated in (Abuhaimed *et al* 2014).

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